SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: ITA0606 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO1: *Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)*

Assessment Tool Weightage: 20%

QUESTIONS: 15 **Total Marks: 25**

Annexure A

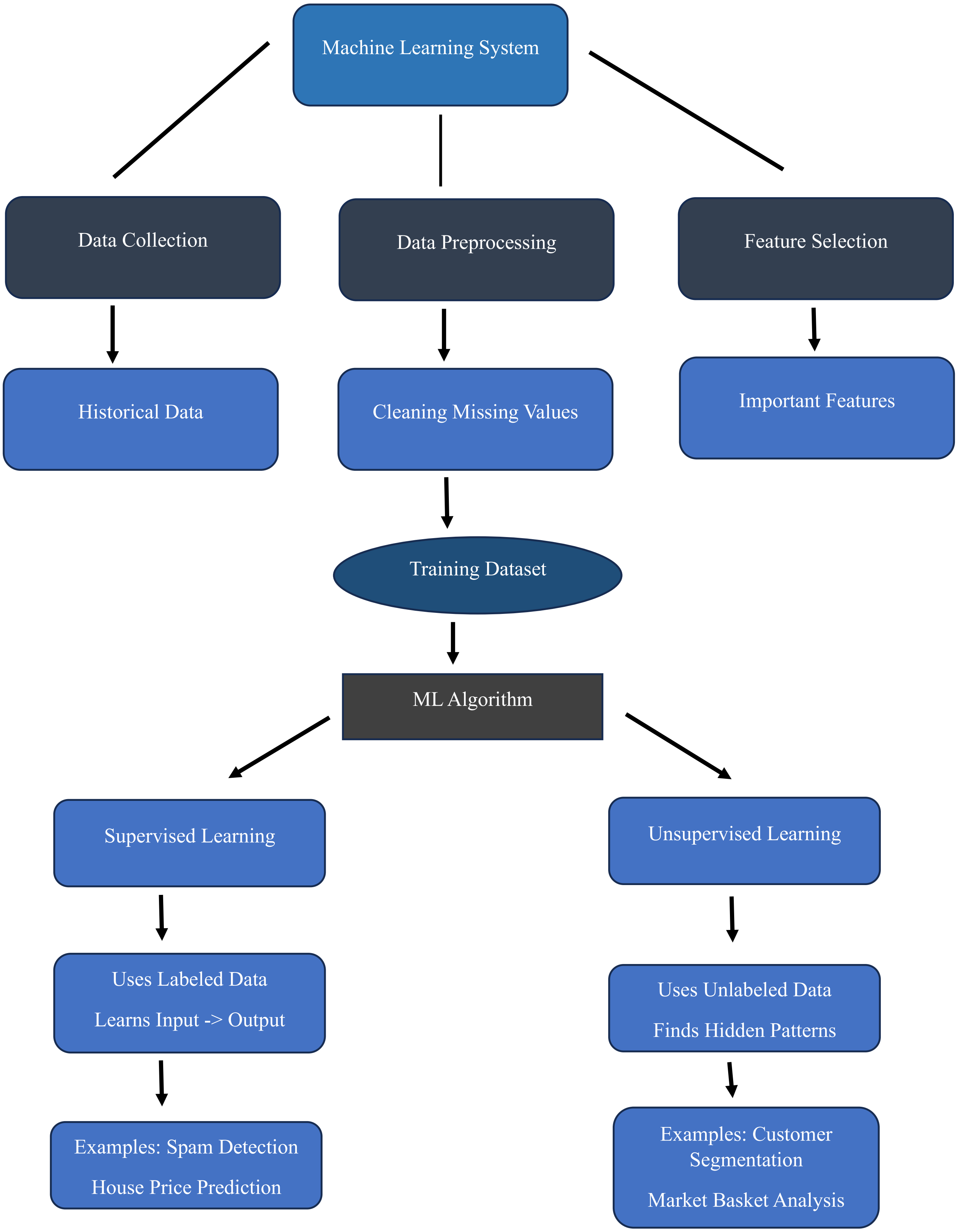
CONCEPT MAPPING

Code of Conduct:

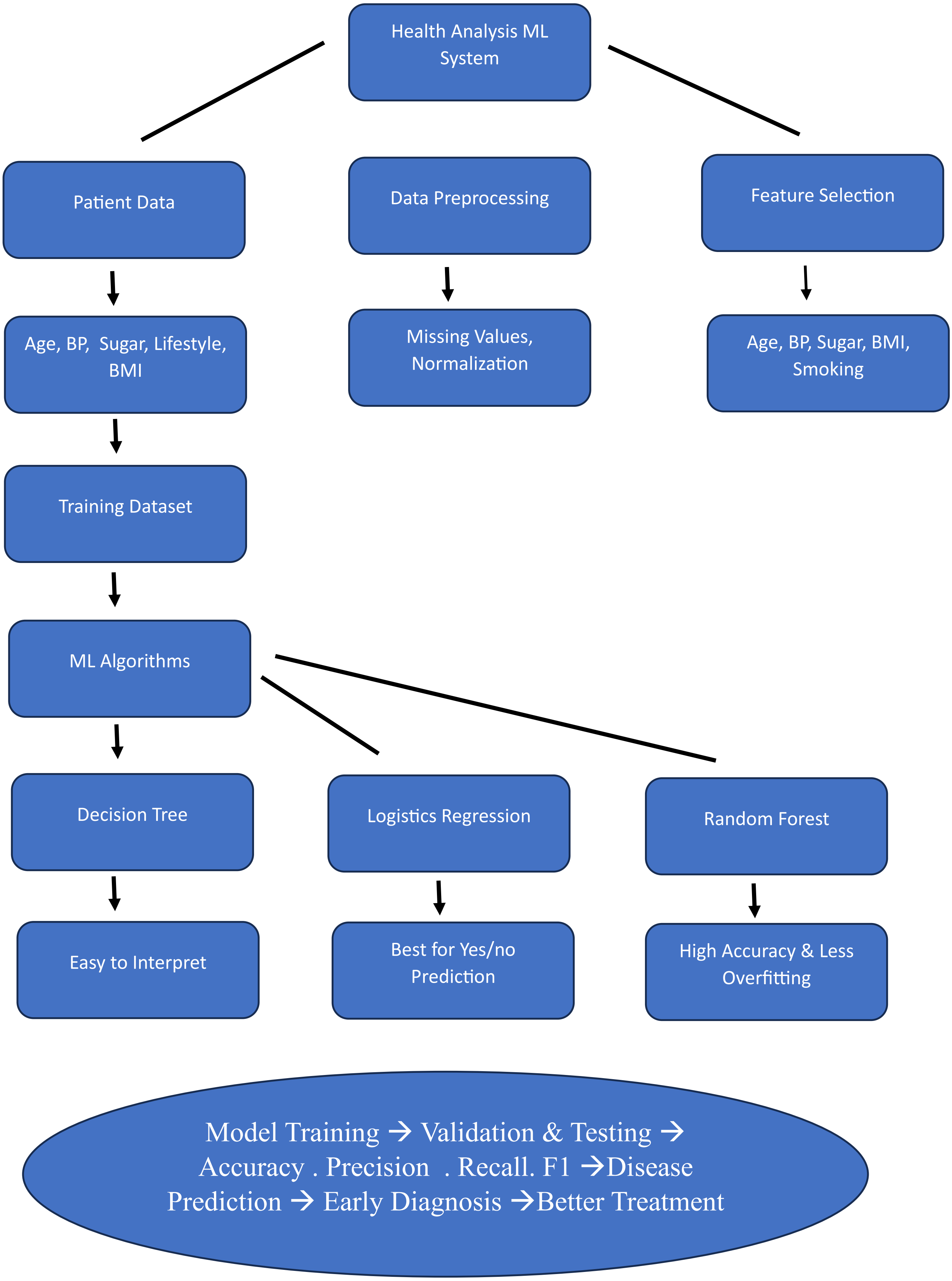
I S. Dharani Teja certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
S. Dharani Teja

1. Machine Learning System, Supervised & Unsupervised Learning

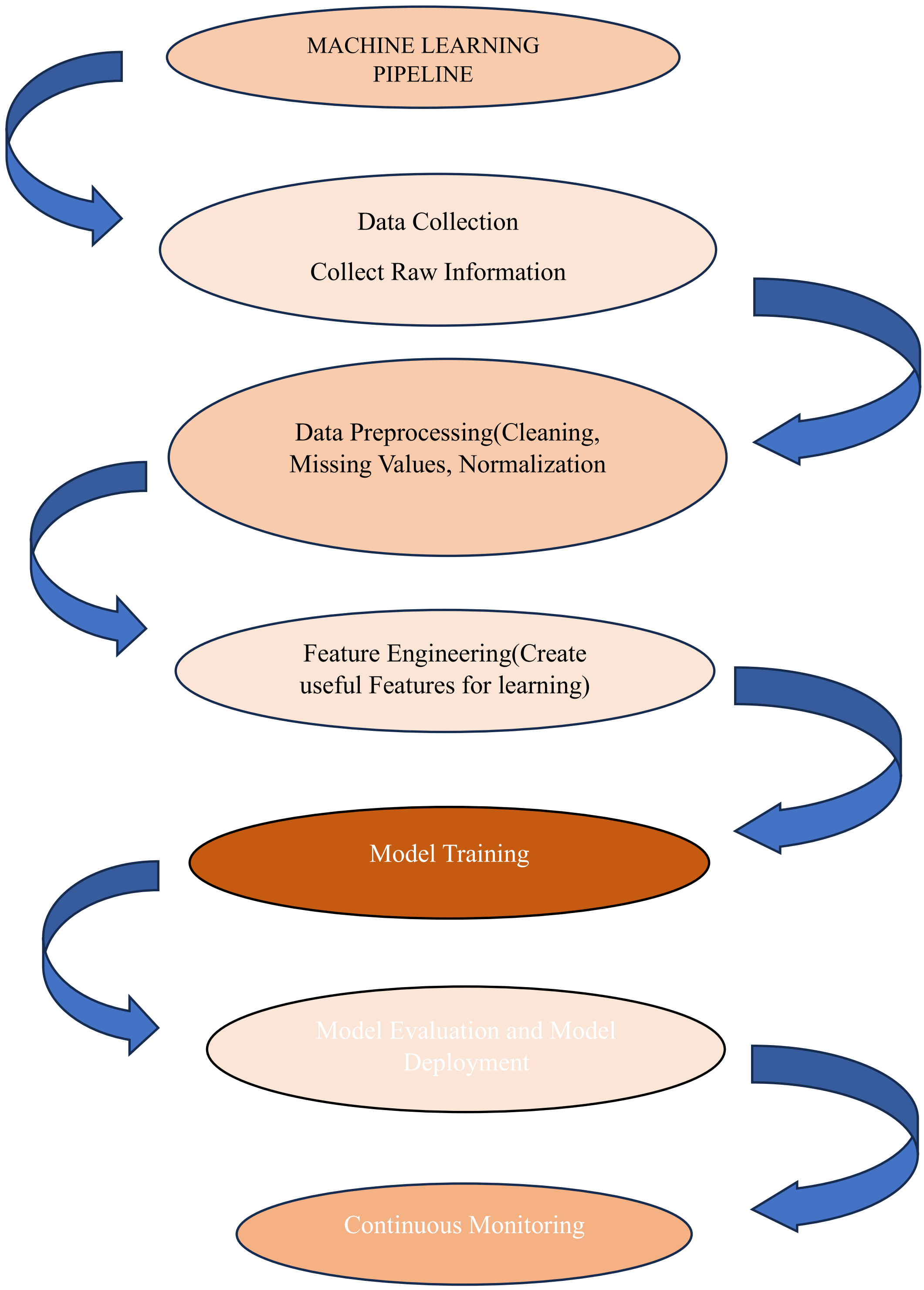


10. Machine Learning System for Health Analysis



- **Decision Tree:** Easy to understand by doctors.
- **Logistic Regression:** Best for binary prediction (Disease/No Disease).
- **Random Forest:** High accuracy and reduces overfitting.

11. Machine Learning Pipeline



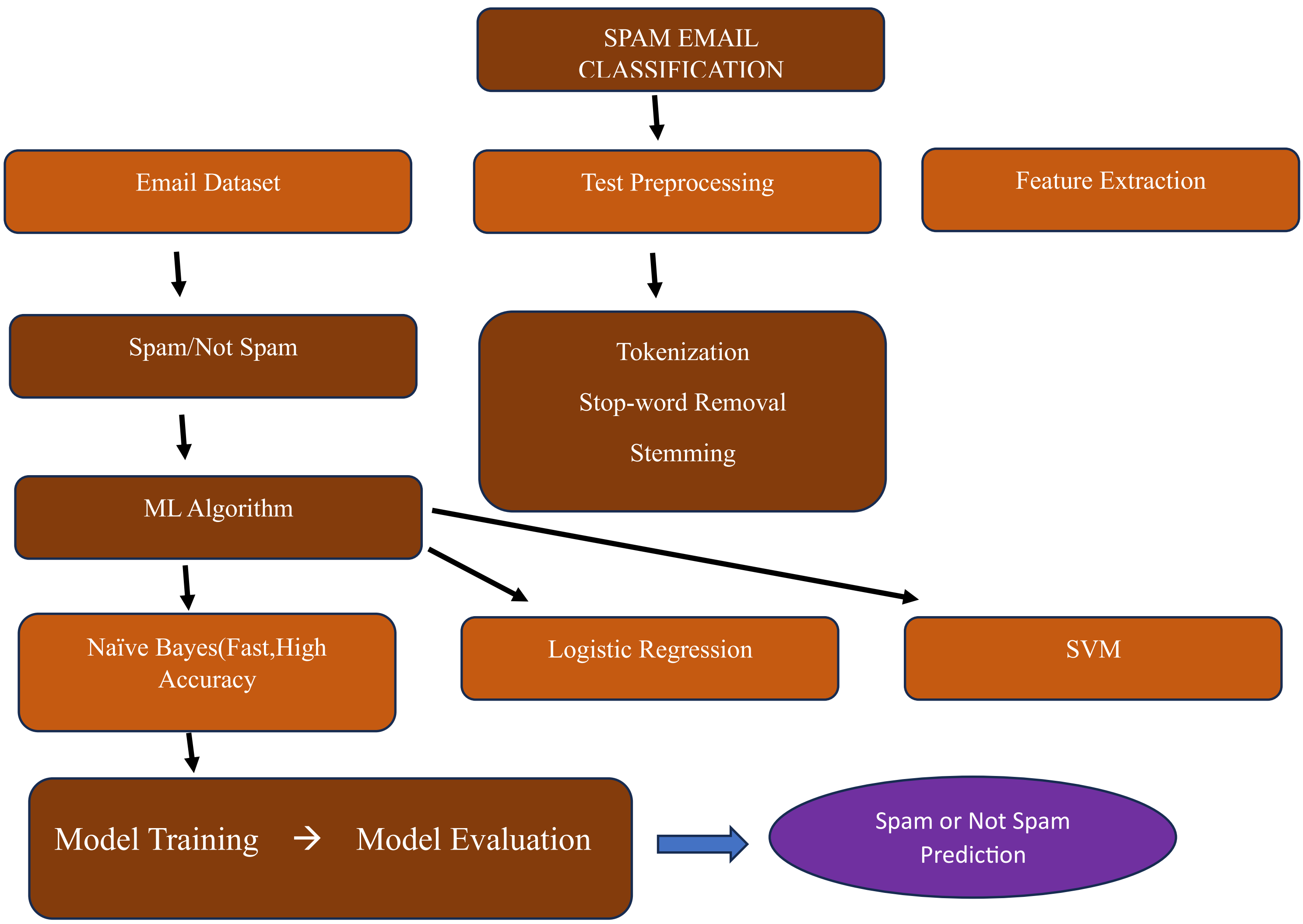
Training Error vs Generalization Error

Training Error	**Generalization Error**
Error on training data	Error on unseen test data
Used during learning	Measures real-world performance
Lower is better	Lower indicates good generalization

Example:

- Training Accuracy = **99%**
- Testing Accuracy = **90%**
- Difference indicates **overfitting**.

12. Spam Email Classification



15.Explain the concept of clustering in unsupervised learning.

Clustering in Unsupervised Learning

- Clustering is an **unsupervised machine learning technique** used to group similar data points into clusters.
- It works on **unlabeled data**, meaning the output categories are not predefined.
- Data points within the same cluster are more similar to each other than to those in other clusters.
- The main objective is to discover hidden patterns or relationships in the data.

(a) Working of the K-Means Algorithm

Steps of the K-Means Algorithm

1. Choose the number of clusters (**K**).
2. Randomly select **K centroids** (initial cluster centers).
3. Calculate the distance between each data point and every centroid.
4. Assign each data point to the nearest centroid.
5. Calculate new centroids by finding the mean of all points in each cluster.
6. Repeat the assignment and centroid update steps until the centroids no longer change or the maximum number of iterations is reached.
7. The final clusters are formed when the algorithm converges.

Example

Suppose a shopping mall wants to group customers based on **Annual Income** and **Spending Score**.

- Select **K = 2**.
- Randomly choose two centroids.
- Assign each customer to the nearest centroid.
- Update the centroid positions.

- Repeat the process until the centroids become stable.
- **Final Output:**
  - Cluster 1 – High-income, high-spending customers.
  - Cluster 2 – Low-income, low-spending customers.

(b) Elbow Method

- The **Elbow Method** is used to determine the optimal value of **K**.
- Run the K-Means algorithm for different values of **K** (e.g., 1 to 10).
- Calculate the **Within-Cluster Sum of Squares (WCSS)** for each value of K.
- Plot a graph with:
  - **X-axis:** Number of clusters (K)
  - **Y-axis:** WCSS
- As K increases, WCSS decreases.
- The point where the graph forms an **"elbow"** is considered the optimal value of K.
- Choosing K at the elbow provides a good balance between accuracy and complexity.

(c) Comparison between K-Means and Hierarchical Clustering

K-Means	Hierarchical Clustering
Requires the value of **K** before clustering.	Does not require specifying **K** initially.
Uses **centroids** to create clusters.	Uses a **tree structure (dendrogram)** to form clusters.
Faster and suitable for large datasets.	Slower and suitable for small datasets.
Produces flat (non-hierarchical) clusters.	Produces hierarchical (nested) clusters.

K-Means	Hierarchical Clustering
Sensitive to the initial centroid selection.	Less dependent on initialization.

Conclusion

- Clustering is an important unsupervised learning technique used to group similar data.
- **K-Means** is simple, fast, and efficient for large datasets.
- The **Elbow Method** helps select the optimal number of clusters.
- **Hierarchical Clustering** is useful when the relationship between clusters needs to be visualized.
- The choice of clustering algorithm depends on the dataset size and application requirements.